ELEC205 Assignment 1

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Question 1

1. Truth Table and Conversion

First, the given function in maxterm form is converted to minterm form. Since maxterms represent the rows where the output is 0, the remaining rows will have an output of 1:

The corresponding truth table was filled in logisim as shown in figure 1:

16 of 16 rows shown

A	B	C	D	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

Figure 1

2. K-Map Minimization

Using the truth table, the Karnaugh Map is generated to find the simplified Sum-of-Products expression. The groups are formed as shown in figure 2:

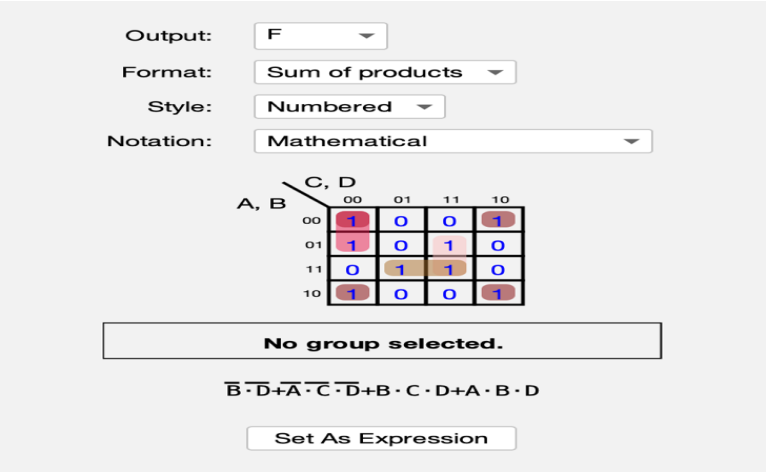


Figure 2

3. Circuit Implementation

The simplified expression is implemented in Logisim using standard logic gates. The final circuit design is shown in figure 3:

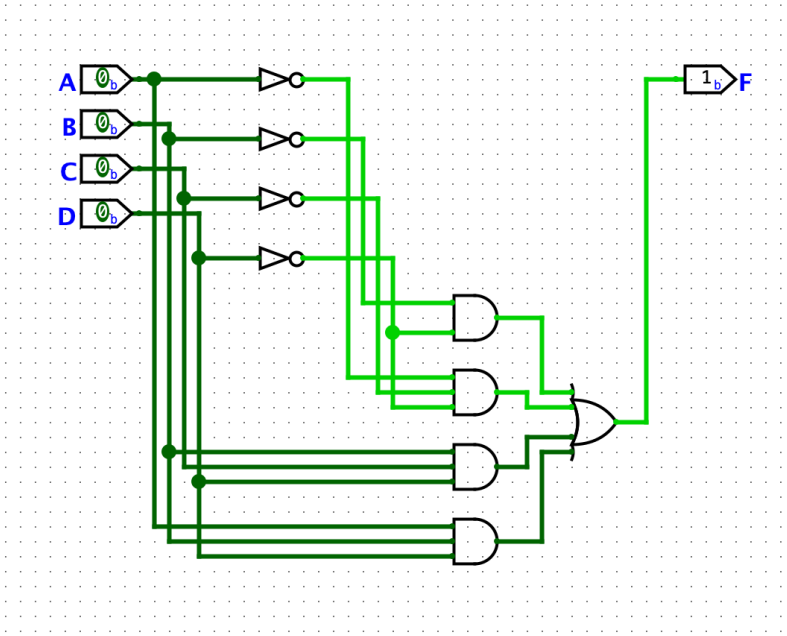


Figure 3

Question 2 (a)

1. Circuit Implementation (Manual Design)

The circuit was manually designed using AND, OR, and NOT gates according to the given expression. Each product term was implemented with AND gates and combined using a 4-input OR gate.

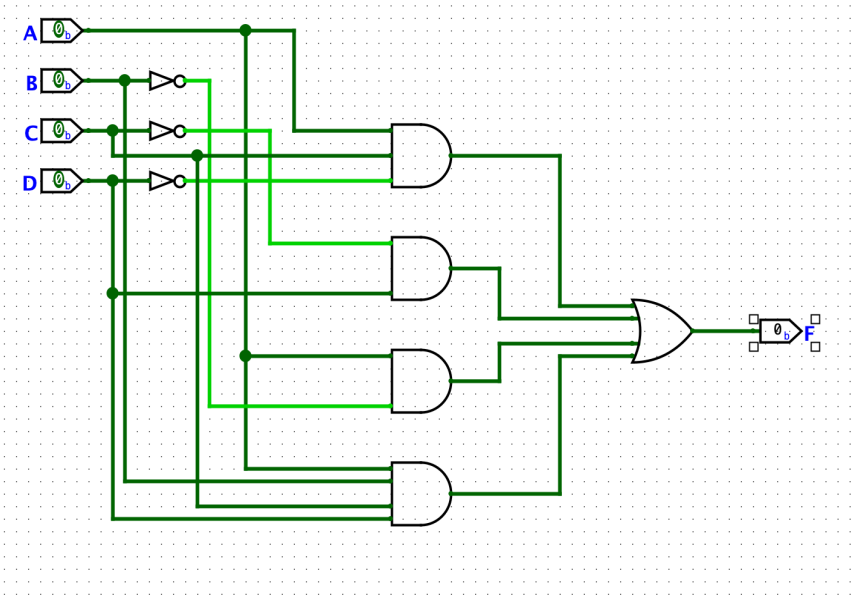


Figure 4

2. Truth Table

After implementing the circuit, the truth table was extracted using the analyze circuit tool. The table shows the output for all 16 possible input combinations.

16 of 16 rows shown

A	B	C	D	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Figure 5

3. Karnaugh Map and Simplification

The Karnaugh Map was generated to simplify the original expression. Both SOP and POS forms were investigated:

Sum-of-Products Form:

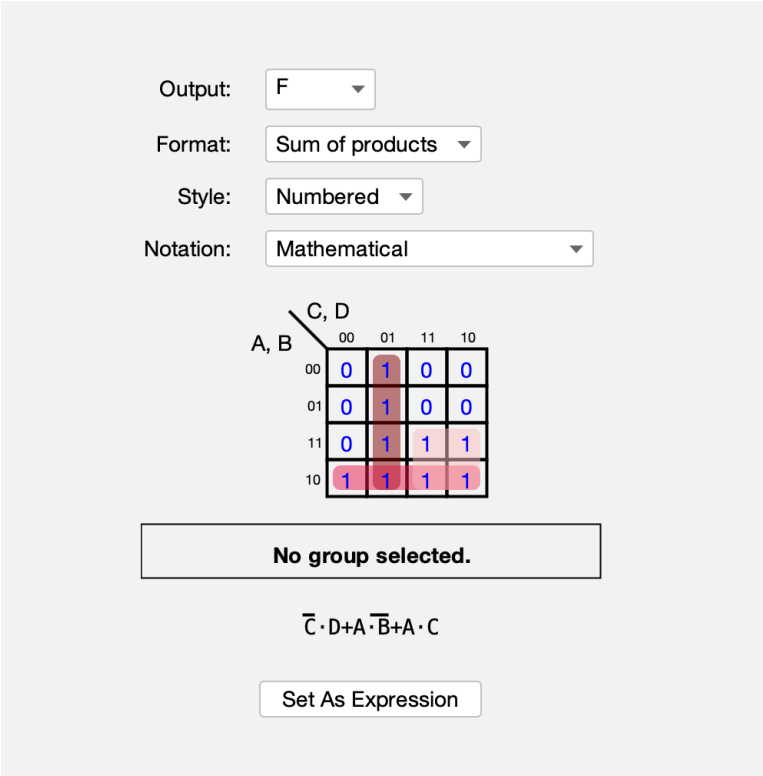


Figure 6

Product-of-Sums (POS) Form:

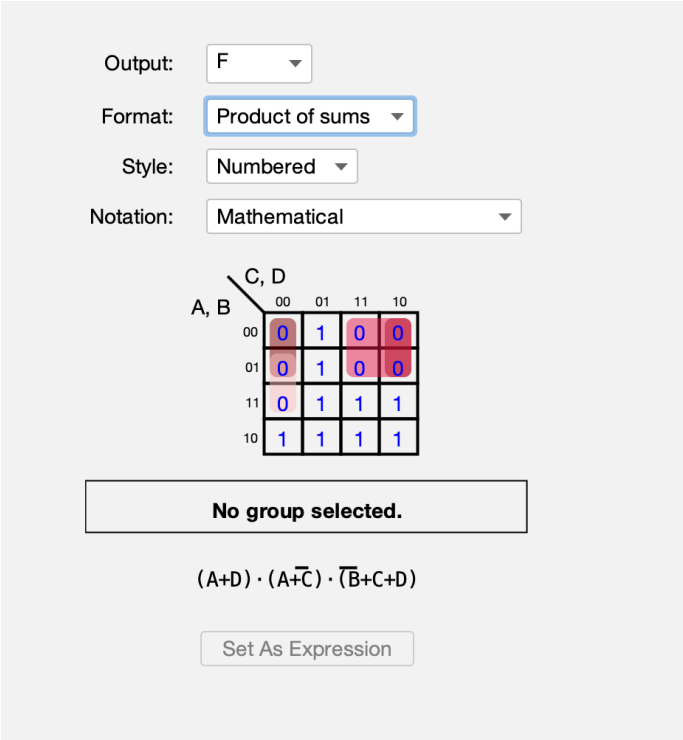


Figure 7

Question 2 (b)

1. Circuit Implementation

The circuit was manually designed using a Product-of-Sums (POS) approach. Each clause in parentheses was implemented using 3-input OR gates, and their outputs were combined using a 4-input AND gate to produce the final output .

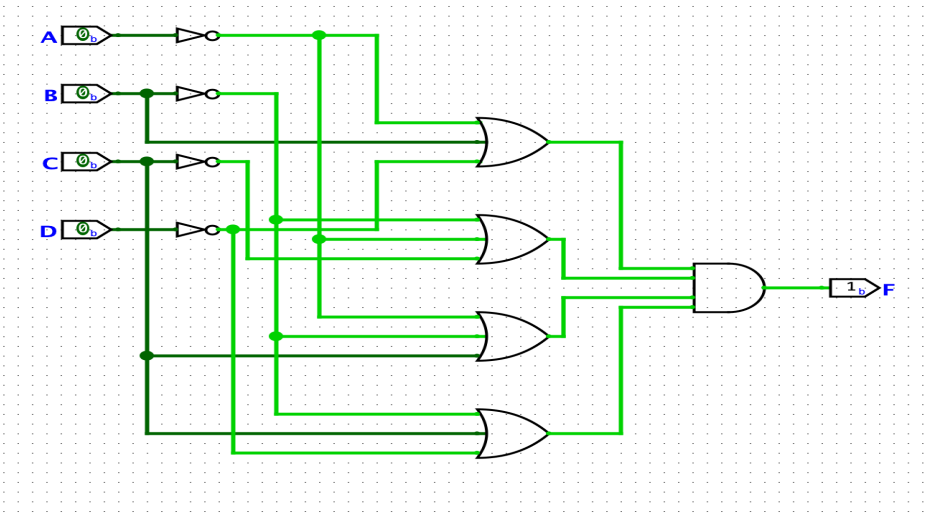


Figure 8

2. Truth Table

The truth table for the 4-variable function was extracted. It defines the logic state of for all possible combinations of and .

16 of 16 rows shown

A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Figure 9

3. Karnaugh Map and Simplification

The function was analyzed via Karnaugh Map to identify the most efficient simplified forms:

Sum-of-Products Form:

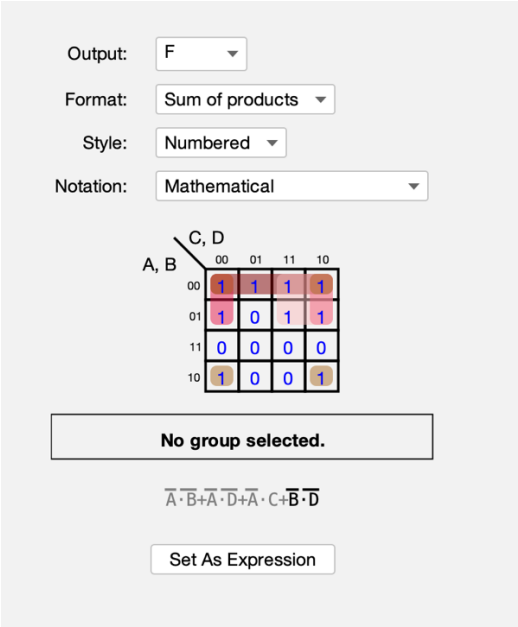


Figure 10

Product-of-Sums (POS) Form:

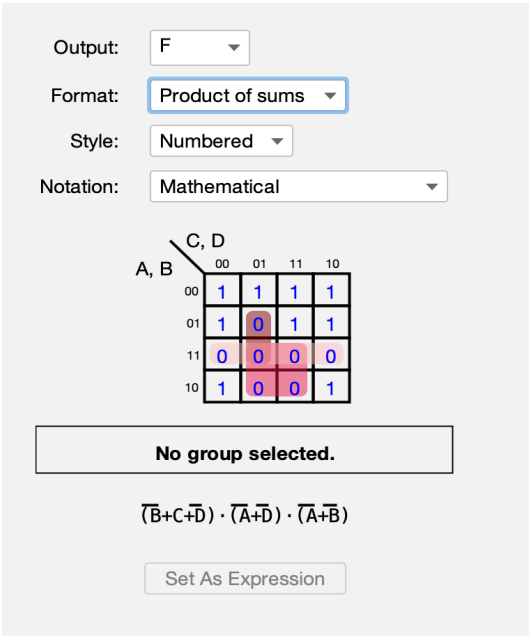


Figure 11

Question 3

1. Integrated Circuit Design

- **Top Circuit (Output F):** Implemented using only NAND gates using Logisim's build circuit feature.
- **Bottom Circuit (Output G):** Manually designed using only NOR gates based on the Product-of-Sums structure.

- **Common Inputs:** Both circuits share the same input pins to ensure synchronized testing.

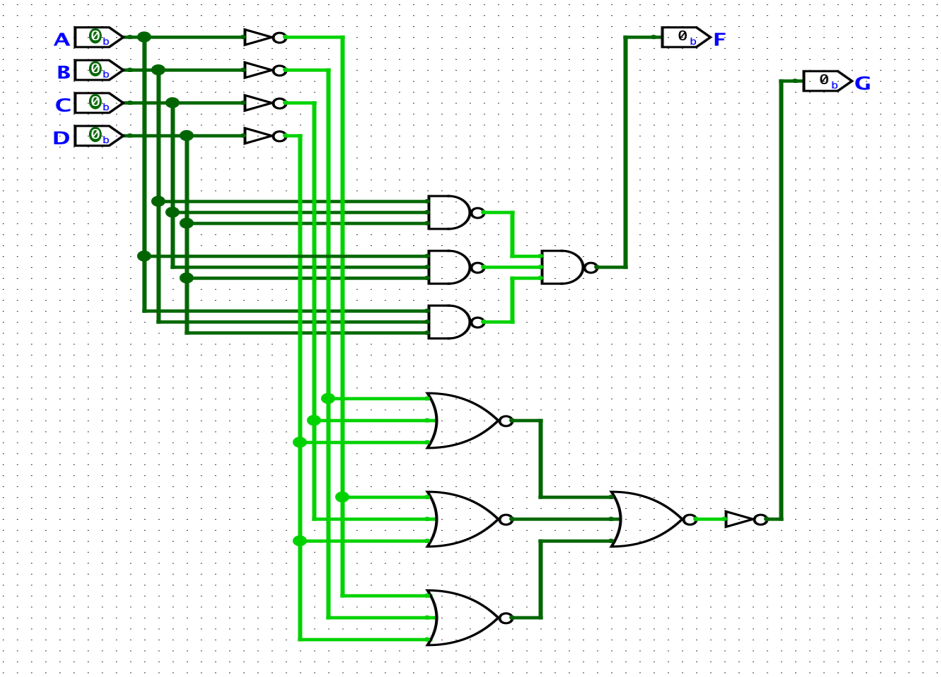


Figure 12

2. Combined Truth Table Verification

The truth table below displays the outputs for both F and G. As observed, the F and G columns are identical for all 16 input combinations, proving that both circuits are logically equivalent.

16 of 16 rows shown

A	B	C	D	F	G
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	0	0	0
0	0	1	1	0	0
0	1	0	0	0	0
0	1	0	1	0	0
0	1	1	0	0	0
0	1	1	1	1	1
1	0	0	0	0	0
1	0	0	1	0	0
1	0	1	0	0	0
1	0	1	1	1	1
1	1	0	0	0	0
1	1	0	1	1	1
1	1	1	0	0	0
1	1	1	1	1	1

Figure 13

3. Timing Diagram Analysis

To further verify the dynamic behavior of the circuits, a Timing Diagram was generated. The waveforms for F and G perfectly align with each other throughout the simulation, confirming that they react identically to changes in inputs.

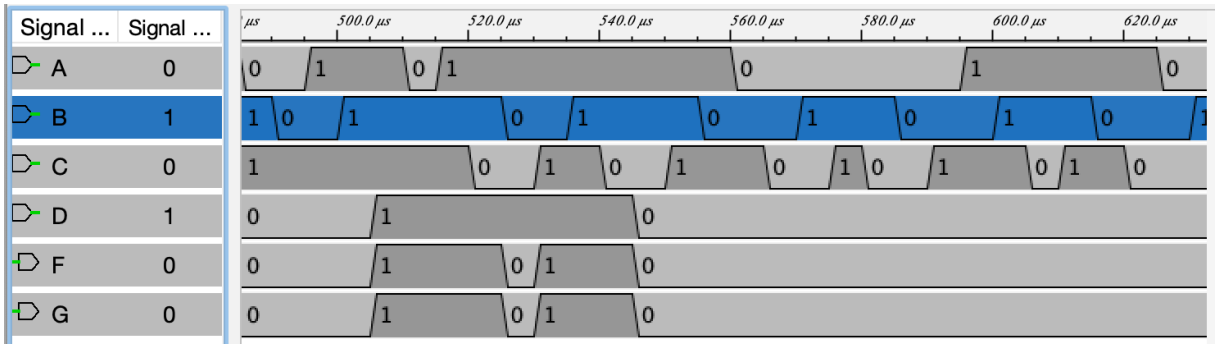


Figure 14

4. K-Map Minimization for F and G

The Karnaugh Maps for both outputs confirm that they share the same minimized Boolean expression.

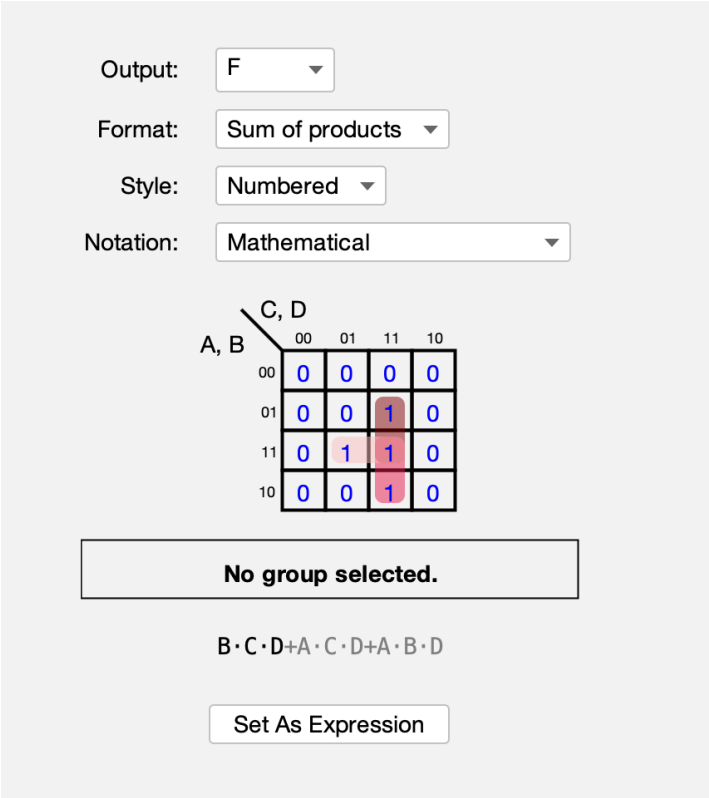


Figure 15

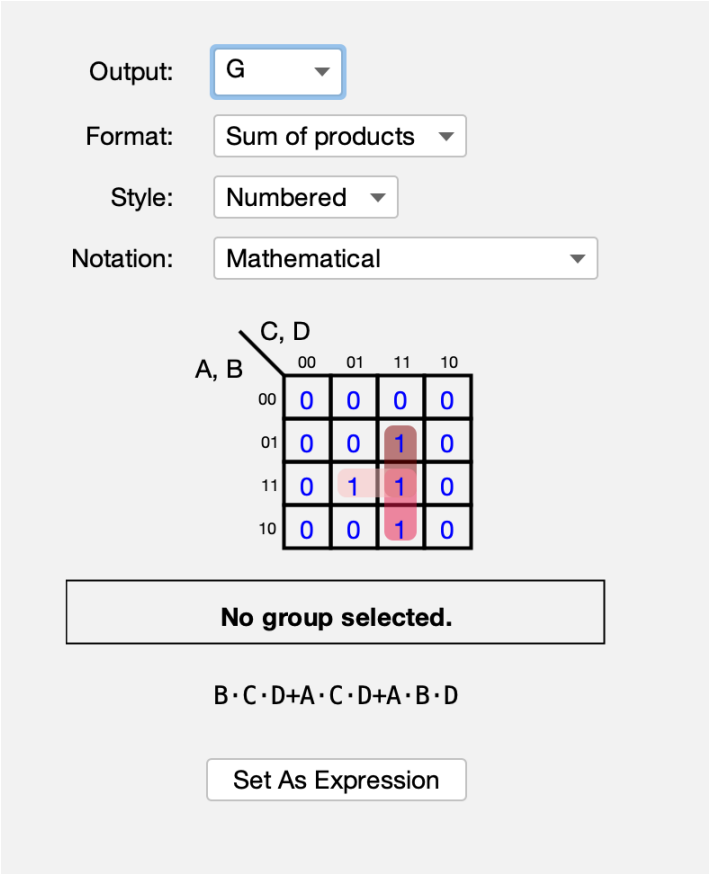


Figure 16

Design Problem

1. Truth Table and Logic Formulation

To design the circuit, a truth table with 16 rows was constructed. The inputs AB and CD represent decimal values from 0 to 3 . The output XY was calculated for each case

16 of 16 rows shown

A	B	C	D	X	Y
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	1
0	1	0	0	0	1
0	1	0	1	0	0
0	1	1	0	0	1
0	1	1	1	1	0
1	0	0	0	1	0
1	0	0	1	0	1
1	0	1	0	0	0
1	0	1	1	0	1
1	1	0	0	1	1
1	1	0	1	1	0
1	1	1	0	0	1
1	1	1	1	0	0

Figure 17

2. K-Map Minimization

The truth table was used to create Karnaugh Maps for each output bit to determine the minimal Sum-of-Products expressions.

Minimization for X (MSB):

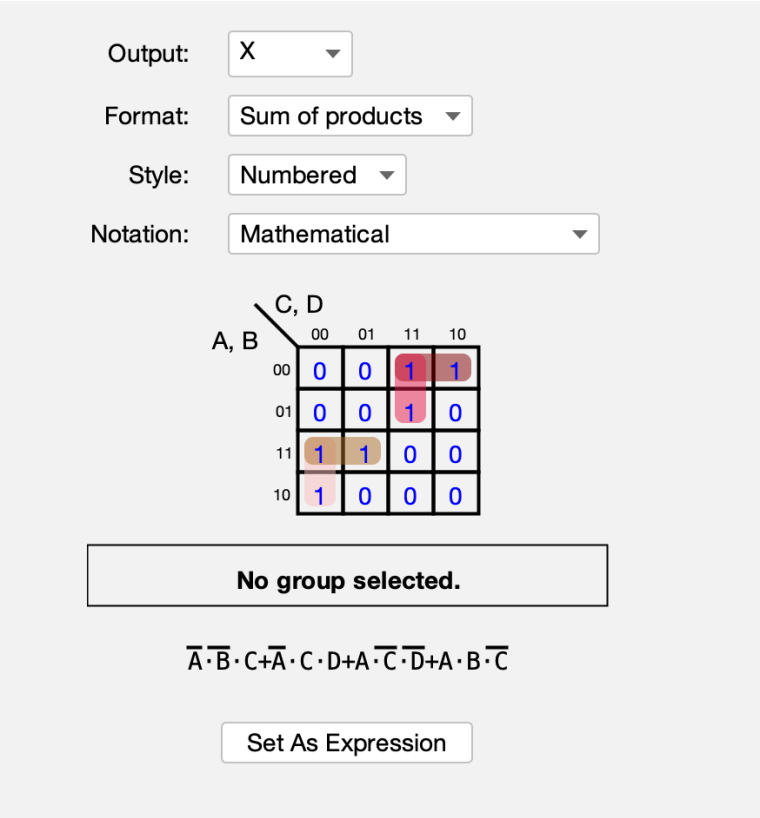


Figure 18

Minimization for Y (LSB):

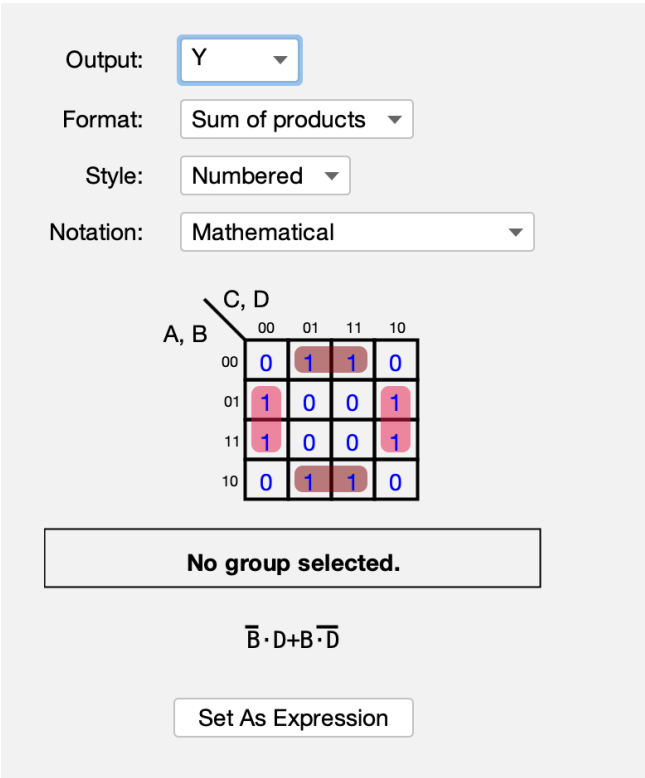


Figure 19

3. Schematic Entry

The minimized Boolean functions were implemented in logisim. The circuit uses AND and OR gates to represent the SOP logic, with NOT gates providing the complemented inputs.

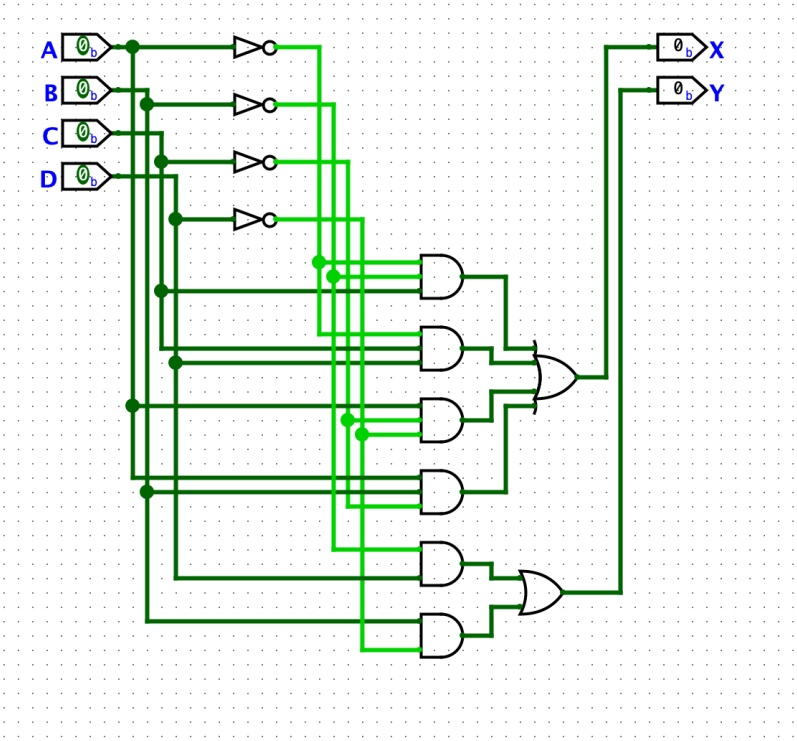


Figure 20

4. Verification and Timing Analysis

The correctness of the design was verified using the Timing Diagram tool. By changing the 4-bit input combinations, the output signals X and Y were observed to follow the expected absolute difference values accurately over time.

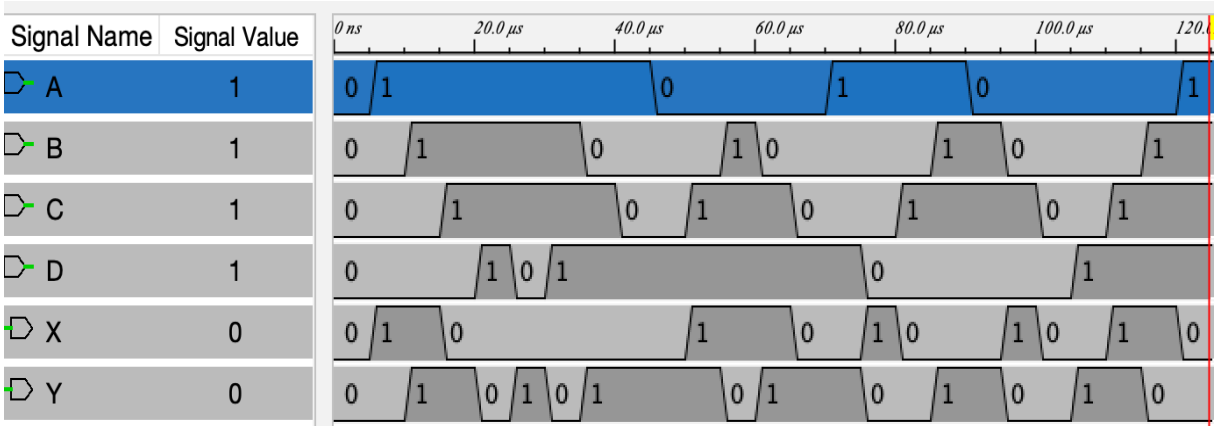

Signal Name
Signal Value
A
1
B
1
C
1
D
1
X
0
Y
0
0 ns
20.0 µs
40.0 µs
60.0 µs
80.0 µs
100.0 µs

Figure 21